

Mika Be

APPLIED AI | SOFTWARE SYSTEMS | COMPUTER ENGINEERING

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EDUCATION

Rutgers University - New Brunswick Expected May 2028
B.S. in Electrical and Computer Engineering | Minor in Computer Science
GPA 3.838/4.000 | Dean's List, all four completed semesters
Coursework: Data Structures; Digital Logic Design; Computer Architecture & Assembly (current); Linear Systems & Signals (current)

HONORS & AFFILIATIONS

IEEE-HKN (Eta Kappa Nu), Gamma Epsilon Chapter | Member 2026-Present
Bergen County Academies - Computer Science Departmental Award | 1 of 3 recipients 2024

RESEARCH

Kuiper TTS | Undergraduate Researcher & Co-author 2025-Present
Python | *PyTorch Lightning* | *Piper/VITS* | *ONNX* | *LaTeX*
- Led an 18-member interdisciplinary team through manuscript preparation and reproducibility work; co-authored 'Making Legacy Scientific Papers Accessible by Converting Equations into Spoken Language,' which received the 2nd Place Best Full Paper Award at IEEE ISEC 2026.
- Built a reproducible GPU training and ONNX export pipeline for Piper/VITS voices; repaired Python and dependency incompatibilities and validated cross-platform inference.
- Engineered LaTeX-to-speech transformations for fractions, derivatives, grouping, and operator-aware pacing, with traceable manifests linking source text, settings, audio, and duration.

SELECTED PROJECTS

Room Notes | Solo Developer | Private beta 2026-Present
Flutter | *Dart* | *Firebase* | *Drift/SQLite*
- Built a cross-device spatial notebook where text, ink, equations, code, images, and diagrams share a consistent 900 x 1200 page model.
- Designed local-first persistence and conflict-aware synchronization for notebooks, media, revisions, and shared lectures, with automated regression coverage.

Knowsy | Technical Lead | 3-person team 2025-Present
JavaScript | *circuit simulation* | *automated testing*
- Led development of an interactive breadboard lab with wiring validation, digital simulation, and inspectable netlist output.
- Implemented modified nodal analysis over complex admittances and frequency sweeps for voltage, phase, and Bode-plot calculations.

Smart Plant Care Robot | Embedded Systems Project 2024
ESP32 | *sensor integration* | *transistor-switched loads* | *power validation*
- Built a monitoring and irrigation prototype with soil-moisture, light, and environmental sensors; drove pumps through flyback-protected circuits and validated 5-12 V power using multimeter and serial diagnostics.

Room of Days | Solo Developer | Active development 2025-Present
Flutter | *Dart* | *local-first architecture* | *Firebase*
- Built a planning and reflection app for Android, iOS, and web with an offline core, validated JSON persistence, backups, and optional cloud sync.

EXPERIENCE

Schieber Research Group | Software Developer 2023-2024
React | *Vite* | *Appwrite*
- Built and documented a full-stack research prototype, translating study requirements into reusable interfaces and a maintainable web application.

Starbucks | Barista 2023-2025

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, Dart, SQL, HTML/CSS
AI & research: PyTorch Lightning, Piper/VITS, ONNX, GPU training, LaTeX parsing, experiment reproducibility
Software: Flutter, Firebase, Drift/SQLite, React, Git, automated testing, local-first synchronization
Hardware & systems: Circuit simulation, digital logic, Arduino/ESP32, LTspice, breadboarding, soldering